SMART INDIA HACKATHON 2026

PanchayatCast

Complete Data Strategy & Acquisition Guide

Hybrid statistical + ML downscaling of weather forecasts from Block level to Panchayat level

PROBLEM STATEMENT ID

26074

TITLE

Downscaling of weather forecast from Block level to Panchayat level

ORGANIZATION

Ministry of Earth Sciences (MoES) - India Meteorological Department

CATEGORY

Software

THEME

Agriculture, FoodTech & Rural Development

Access terms, quotas and URLs change often - verify each source before the hackathon.

At a Glance: What to Use in 36 Hours

Short answer: use **Open-Meteo** as the live forecast feed (formatted to look exactly like an IMD block forecast), **ERA5 / ERA5-Land** + **CHIRPS** + **IMD gridded** data for training, **Copernicus/SRTM DEM** + **ESRI 10 m LULC** as static predictors, and **state AWS data** for validation. Keep IMD's official API as the production adapter you *show*, and NASA POWER / Visual Crossing as backups.

Role in PanchayatCast	Primary (use in demo)	Backup
Live 5-day block forecast input	Open-Meteo forecast API (GFS / ECMWF models)	IMD API (if whitelisted), Visual Crossing, WeatherAPI
Coarse training field ("block scale")	ERA5 0.25° / IMD gridded rain 0.25°, temp 1°	NASA POWER
Fine training target ("panchayat scale")	ERA5-Land 0.1° (temp, RH, wind); CHIRPS 0.05° (rain)	IMERG (rain), MODIS LST (temp)
Static predictors	Copernicus GLO-30 / SRTM DEM; ESRI 10 m LULC; water distance	CartoDEM, Bhuvan LULC, Dynamic World
Boundaries	State GIS / Bhuvan GP layer + LGD codes	OSM village points -> Voronoi polygons
Validation truth	State AWS network (e.g. TGDPS Telangana)	Meteostat / NOAA GSOD, IMD station data

2. Primary Weather Data Sources

2.1 IMD Open Data / API

BACKUP

What it provides	Official IMD observation and forecast products: current weather, district/city forecasts, nowcasts, warnings, AWS/ARG readings (product list is in IMD's API documentation).
Resolution	Station, city and district level; block-level agromet forecasts are published via GKMS bulletins.
How to access	IMD lists its APIs at mausam.imd.gov.in (API documentation + IP-whitelisting request). Access is granted to organisations on request, for whitelisted server IPs, under IMD terms.
Advantages	The official, authoritative source - exactly what the final product would consume.
Limitations	Whitelisting takes time and is unlikely to be approved within a hackathon; hackathon Wi-Fi IPs change. **Strategy:** build an *IMD adapter* with the same schema and show it in the architecture, but feed the demo from Open-Meteo or cached samples.

2.2 IMD Gridded Data (IMD Pune)

RECOMMENDED - USE IN DEMO

What it provides	Long-term daily gridded rainfall (from 1901) and daily max/min temperature (from 1951) built from IMD's station network.
Resolution	Rainfall 0.25° x 0.25° (~25 km); temperature 1° x 1° (~100 km). Daily.
How to access	Free download from the IMD Pune data portal (binary .grd files). Easiest route: the **imdlib** Python package (pip install imdlib), which downloads and converts to xarray/NetCDF.
Advantages	Indian ground-truth-based, trusted by IMD judges, long record for climatology and model training.
Limitations	Too coarse to be the fine-scale target; historical only (no forecasts); temperature grid is very coarse. Use it as the **coarse input** and for climatology, not as panchayat truth.

2.3 Open Government Data Platform (data.gov.in)

OPTIONAL / LATER

What it provides	Published IMD datasets such as sub-division/district rainfall statistics, rainfall normals, and assorted agriculture and village-level datasets.
Resolution	Mostly district / sub-division, monthly or annual aggregates.
How to access	Web download (CSV/XLS) or API with a free data.gov.in API key.
Advantages	Official, free, easy CSVs; useful for district rainfall normals and context slides.
Limitations	Not daily gridded data; too coarse for downscaling. Use for background statistics only.

2.4 NASA POWER

BACKUP

What it provides	Daily/hourly temperature, humidity, precipitation, wind and solar radiation for any point on Earth, from 1981 onward (agroclimatology community).
Resolution	About 0.5° x 0.625° for meteorology (MERRA-2 based); solar parameters on a coarser grid.
How to access	Free REST API (no key): power.larc.nasa.gov - query by lat/lon and date range; returns JSON/CSV.
Advantages	Extremely easy, reliable, includes agriculture-friendly variables (solar radiation, ET inputs).
Limitations	Coarse; reanalysis-based precipitation is weak for Indian monsoon convection; no multi-day forecast.

2.5 Open-Meteo

RECOMMENDED - USE IN DEMO

What it provides	Forecast API (up to 16 days, hourly/daily) from multiple global models (e.g. GFS, ECMWF IFS, ICON), plus a Historical Weather API backed by ERA5 / ERA5-Land and an Elevation API.
Resolution	Depends on model selected (roughly 9-25 km globally); historical ERA5-Land about 9 km.
How to access	Free, no API key for non-commercial use within fair-use daily limits; simple HTTP GET with lat/lon. Batch many coordinates in one call.
Advantages	**Best choice for the live demo:** real, current multi-model forecasts in minutes; same API for history and forecast; model choice lets you show ensemble spread for the confidence badge.
Limitations	Not an official IMD product - say so openly. Check current usage terms; cache responses so rate limits cannot break the demo.

2.6 ERA5 / ERA5-Land (Copernicus CDS API) and NOAA GFS

RECOMMENDED - USE IN DEMO

What it provides	ERA5: global hourly reanalysis (1940-present) of all key surface variables. ERA5-Land: land-surface re-run at higher resolution incl. 2 m temperature, dew point, wind, soil moisture. NOAA GFS: operational global forecast (open).
Resolution	ERA5 0.25° (~28 km); ERA5-Land 0.1° (~9 km); GFS 0.25°.
How to access	Register on the Copernicus Climate Data Store, accept the dataset licence, save your API token, then use the **cdsapi** Python client. GFS via NOAA NOMADS or AWS Open Data.
Advantages	Physically consistent, gap-free, long record: ideal **coarse -> fine training pairs** (ERA5 -> ERA5-Land) for perfect-prognosis training.
Limitations	CDS requests are queued - start downloads on Day 0, not during the hackathon. Reanalysis rain is smooth; use CHIRPS/IMERG for rainfall targets. Large files - subset to your district bbox.

2.7 Visual Crossing / WeatherAPI (demo only)

BACKUP

What it provides	Commercial point forecasts and history (temperature, rain, humidity, wind) via simple REST.
Resolution	Point queries; underlying models blended by the provider.
How to access	Free developer key with a daily quota (e.g. Visual Crossing has offered around 1,000 records/day on its free plan; check current plans for both).
Advantages	Very quick to integrate; good fallback if Open-Meteo is unreachable.
Limitations	Quotas, licence restrictions, opaque methods; not suitable for training or a government product.

2.8 Historical Station Data

RECOMMENDED - USE IN DEMO

What it provides	Point observations for **validation**: daily rain, Tmax/Tmin, RH, wind.
Sources	**State AWS networks** (e.g. Telangana TGDPS, Karnataka KSNDMC - dense, often published publicly); **IMD station data** (IMD Pune data supply portal, on request, may take time); **Meteostat** and **NOAA GSOD** (open, fewer Indian stations).
How to access	Download daily CSVs from state portals; meteostat Python library; GSOD from NOAA.
Advantages	The only true ground truth - essential for the accuracy slide judges will ask about.
Limitations	Gaps and quality issues; sparse in some districts. Always hold stations **out of training** so validation is honest.

3. Static High-Resolution Predictors

Static layers explain *why* weather differs inside a block. They are downloaded once, resampled to a common 1 km grid, and reused every day.

Predictor	Source & resolution	Why it helps downscaling	Priority
Elevation, slope, aspect, TPI	SRTM or Copernicus GLO-30 (30 m, OpenTopography / USGS / AWS); CartoDEM 30 m (ISRO Bhuvan, registration)	Temperature falls ~6.5 °C per km of height; slopes and ridges change rain (orographic lift) and wind exposure	**Must**
Land use / land cover	ESRI Land Cover 10 m (annual, free download); Dynamic World 10 m (Google Earth Engine); Bhuvan LULC (ISRO)	Urban areas are warmer, forests and irrigated fields cooler and more humid; also sets surface roughness for wind	**Must**
Vegetation (NDVI)	MODIS 250 m 16-day; Sentinel-2 10 m	Captures seasonal greening and irrigation that affect temperature and humidity	Should
Soil data	ISRIC SoilGrids 250 m (texture, organic carbon); NBSS&LUP soil maps	Useful for the *advisory* (water-holding capacity, irrigation, sowing) more than for the weather model itself	Could
Distance to coast / water bodies	Computed from OSM water polygons, JRC Global Surface Water, coastline	Lakes, tanks and reservoirs moderate temperature and raise humidity; coastal distance matters in coastal districts	**Must**
Latitude & longitude	Panchayat centroid from boundary polygons	Captures smooth regional gradients (e.g. monsoon rainfall gradient) not explained by terrain	**Must**
Climatology layers	Long-term monthly means from CHIRPS / ERA5-Land / WorldClim (1 km)	The single strongest predictor: encodes persistent local patterns ('this GP is always wetter')	**Must**

4. Panchayat Boundary & Administrative Data

Source	What you get	Access	Notes
LGD - Local Government Directory (MoPR)	Official codes and hierarchy: State > District > Block > Gram Panchayat > Villages	lgdirectory.gov.in - downloadable reports (CSV/XLS)	**No geometry**, but LGD codes are the official join key. Store them in every table.
Bhuvan (ISRO/NRSC)	Gram Panchayat boundaries and thematic layers (e.g. Bhuvan Panchayat / SISDP)	Web viewing and WMS services; bulk download may be restricted or need registration	Most authoritative GP geometry; confirm download terms early.
State GIS portals	Village / GP boundaries published by some state remote-sensing or planning agencies	State portal downloads or requests	Often the fastest official route for a single pilot district - check your state first.
GitHub / open repositories	Community boundary files, e.g. DataMeet's India maps (states, districts, some village boundaries)	git clone / direct GeoJSON or shapefile download	Check coverage, vintage and licence; boundaries may be outdated.
Census 2011 / NIC	Village directory, population, village codes; NIC Bharat Maps viewing services	censusindia.gov.in; bharatmaps.gov.in	Use for population/farmer counts and to map villages to GPs via codes.
OpenStreetMap (fallback)	Village point locations (place=village)	Overpass API or Geofabrik extract	If no GP polygons: build Voronoi polygons from village points, clip to block outline.

Tip: decide your pilot district by **boundary availability first**, then check AWS coverage. A district with clean GP polygons and 10+ AWS stations is worth more than a 'more interesting' district without them.

5. Data Handling Strategy

5.1 Handling Missing and Bad Data

Problem	Fix	Do NOT
Station gaps of 1-2 days (temperature, RH)	Linear interpolation in time; flag as filled	Interpolate gaps longer than 2-3 days
Station rainfall gaps	Fill from nearest stations (IDW) or CHIRPS/IMERG for that day; flag	Interpolate rain in time - rain is intermittent
Impossible values (Tmax < Tmin, RH > 100, rain < 0, spikes)	Range + consistency + spike checks; set to missing	Silently keep them in training
Cloud gaps in NDVI	Use 16-day / monthly composites; fill with monthly climatology	Use single cloudy scenes
No-data pixels in DEM / LULC	Nearest-neighbour fill; majority filter for LULC	Leave NaNs - most ML models will fail
API forecast call fails	Serve last cached forecast with a 'stale data' label	Show an empty map in the demo

Keep a **qc_flag** column (0 = raw, 1 = filled, 2 = removed) in every observation table - judges like seeing that data quality was handled deliberately.

5.2 Building a Realistic Demo Dataset if the IMD API Is Restricted

1. **Mimic IMD's format exactly.** Create a block forecast table with the fields used in agromet bulletins: date, day 1-5 rainfall (mm), Tmax, Tmin, RH max/min (%), wind speed (km/h), wind direction, cloud cover (okta).
2. **Fill it from real forecasts.** Query Open-Meteo for a grid of points inside the block, average them to one block value per day - this is a realistic stand-in for the official block forecast.
3. **Use an adapter pattern.** ingest_imd() and ingest_openmeteo() both output the same schema, so swapping to the real IMD API later is one config change.
4. **Record a 'replay' case.** Pick a real past monsoon week with a known local heavy-rain event; store its block forecast, your downscaled output and the AWS observations. This is your guaranteed validation story.
5. **Label it honestly** in the UI: 'Source: Open-Meteo (IMD-format demo feed)'.

5.3 Recommended Sample-Data Strategy for 36 Hours

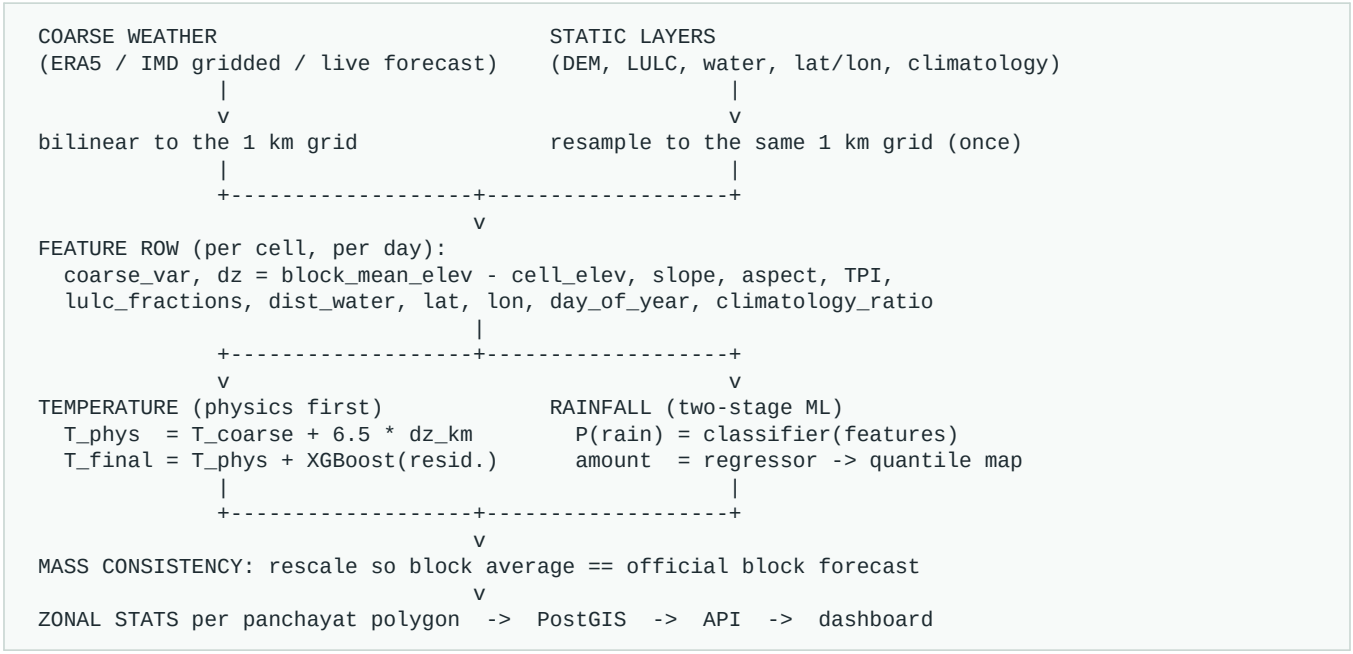
When	Data task	Size target
Before the hackathon (Day -7 to -1)	Submit CDS requests (ERA5, ERA5-Land); download CHIRPS, IMD gridded (imdlib), DEM, LULC, boundaries; collect AWS data; pick the replay event	District bbox only; 10-20 years; < 5 GB total
H0-H6	Clip, reproject (EPSG:32644 / UTM 44N for Telangana), resample all layers to a 1 km grid; build feature stack and climatology	One NetCDF/Zarr feature cube
H6-H16	Create training table (coarse + static -> fine); hold out AWS stations; train models	~1-5 million rows is plenty
H16+	Daily live pull from Open-Meteo; cache every response as JSON	Tiny

6. Recommended Data Pipeline for the Hackathon

6.1 Use in the Demo vs Keep as Backup

Layer	Use in the demo	Keep as backup
Forecast input	**Open-Meteo** (IMD-format block feed)	Cached replay files; Visual Crossing; IMD API adapter
Coarse training	**ERA5 0.25°** + **IMD gridded**	NASA POWER
Fine training target	**ERA5-Land 0.1°** (T, RH, wind); **CHIRPS 0.05°** (rain)	IMERG, MODIS LST
Static predictors	**Copernicus/SRTM DEM, ESRI LULC**, water distance, lat/lon, climatology	CartoDEM, Dynamic World, NDVI, SoilGrids
Boundaries	**State GIS / Bhuvan GP polygons + LGD codes**	OSM village Voronoi polygons
Validation	**State AWS network**	Meteostat / GSOD; IMD stations

6.2 Combining Weather Data and Static Predictors (Hybrid Model)



- **Training (perfect prognosis):** coarse ERA5 / IMD gridded fields are the input, ERA5-Land and CHIRPS the target, over 15-20 monsoon and winter seasons.
- **Inference:** the same features are built from the live block forecast, so the trained model transfers directly.
- **Validation:** compare panchayat output and the 'copy block value' baseline against held-out AWS stations (RMSE/bias for temperature, POD/FAR/CSI for rain).
- **Keep grids aligned:** one CRS, one 1 km grid definition, one bounding box, stored in a config file used by every script.

> **Golden rule for a 36-hour team:** download and clip every historical dataset **before** the hackathon starts. During the 36 hours, the only live network call should be the forecast feed - and even that must be cached.